STAT/MA 41600

In-Class Problem Set #27: October 21, 2026

1. Suppose that X and Y are continuous random variables whose joint probability density function $f_{X,Y}(x,y)$ is constant on the interior of the triangle with vertices at $(0,0)$, $(8,0)$, and $(0,5)$, and otherwise $f_{X,Y}(x,y) = 0$.

1a. For $0 < x < 8$, find the conditional probability density function $f_{Y|X}(y \mid x)$ of Y, given $X = x$.

1b. For $0 < y < 5$, find the conditional probability density function $f_{X|Y}(x \mid y)$ of X, given $Y = y$.

2. Same setup as in question **#1**.

2a. Find the conditional probability that $Y \leq 2$, given $X = 1$. I.e., find $P(Y \leq 2 \mid X = 1)$.

2b. Find the conditional probability that $Y \leq 2$, given $X \leq 1$. I.e., find $P(Y \leq 2 \mid X \leq 1)$.

3. Suppose that X and Y have joint probability density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{4}{1875}(25 - xy) & \text{if } 0 < x < 5 \text{ and } 0 < y < 5, \\ 0 & \text{otherwise.} \end{cases}$$

3a. For $0 < x < 5$, find the conditional probability density function $f_{Y|X}(y \mid x)$ of Y, given $X = x$.

3b. Find the conditional probability that $Y \leq 1$, given $X = 2$. I.e., find $P(Y \leq 1 \mid X = 2)$.

3c. Find the conditional probability that $Y \leq 1$, given $X \leq 2$. I.e., find $P(Y \leq 1 \mid X \leq 2)$.

4. Consider two random variables X and Y with joint probability density function $f_{X,Y}(x,y) = 63e^{-2x-7y}$ for $y > x > 0$, and otherwise $f_{X,Y}(x,y) = 0$.

4a. For $x > 0$, find the conditional probability density function $f_{Y|X}(y \mid x)$ of Y, given $X = x$.

4b. For $y > 0$, find the conditional probability density function $f_{X|Y}(x \mid y)$ of X, given $Y = y$.

5. Same setup as in question **#4**.

5a. Find the conditional probability that $X \leq 3$, given $Y = 5$. I.e., find $P(X \leq 3 \mid Y = 5)$.

5b. Find the conditional probability that $X \leq 3$, given $Y \leq 5$. I.e., find $P(X \leq 3 \mid Y \leq 5)$.